

Algebraic Infinitesimal Probability

Geometric Content, Conditioning, and Certified Precision

Working manuscript accompanying `hyper-lean`

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Abstract

We develop normalized probability and finite-observable integration over the ordered rational-function field $\mathbb{H} = \mathbb{R}(\varepsilon)$, where ε is positive and smaller than every positive real. Exact expressions retain lower-dimensional geometric contributions, while algebraic $\mathcal{O}$ -notation records optional precision. We present checked square, cube, disk, and solid-ball dart contexts, including positive point probabilities, endpoint corrections, and conditioning across dimensions. The geometric content is a rescaled classical Wills polynomial, not a newly discovered valuation. The companion Lean development verifies field arithmetic, normalized contexts, selected geometric identities, and precision rules. It does not yet construct one compatible geometric probability on an unrestricted event domain. This manuscript is a synthesis and formalization draft; novelty and general geometric extensions remain subjects for further review.

1 Purpose and relation to existing work

An exact point can receive positive infinitesimal probability without preventing the whole sample space from having probability one. This idea is established in non-Archimedean probability and numerosity theory [1, 2]. Our scope is narrower: finite observable partitions whose region contents are explicit rational functions. An observable region may represent a continuum of positions; a finite index set is not a claim that the geometric board contains finitely many points.

The proposed contribution is a carefully delimited algebraic interface joining normalization, geometric dimension, exact conditional probability, and certified remainder bounds. None of these ingredients is claimed to be new in isolation. In particular, intrinsic volumes and the Wills polynomial supply the geometric coefficients [3, 4]. A claim that this particular synthesis is original requires a fuller literature review than the initial comparison reported here.

2 The exact field and the meaning of algebraic

Let $\mathbb{H} = \mathbb{R}(\omega)$ be a rational-function field, ordered by declaring the indeterminate ω positive infinite. Equivalently, the sign of a nonzero rational function is the sign of the ratio of its leading polynomial coefficients. Define $\varepsilon = \omega^{-1}$. Then

$$0 < \varepsilon < r \quad (r \in \mathbb{R}, r > 0), \quad \varepsilon\omega = 1.$$

All arithmetic below takes place in this exact ordered field. Inverses of mixed expressions, such as $(1 + \varepsilon)^{-1}$, are retained as rational functions. Finite Laurent polynomials alone would not form a field.

Here “algebraic” means finite field operations, polynomial identities, and order comparisons over specified real geometric data. It does not mean algebraic over $\mathbb{Q}$: π , for example, is an allowed real

coefficient. Nor is ε algebraic over $\mathbb{R}$; it is an indeterminate. The formalization uses noncomputable real coefficients, so a claim of general effective computation with arbitrary real inputs is not made.

Remark 2.1 (Scope of the word hyperreal). This field is a non-Archimedean ordered extension, not a model of full hyperreal transfer. For example, ε has no square root in $\mathbb{R}(\varepsilon)$: the order of a squared rational function at zero is even, whereas that of ε is one. Real closure would be a further algebraic extension. No transfer principle, hyperfinite index set, or infinite summation is assumed in the present probability constructions.

3 Normalized observable contexts

Definition 3.1. A context consists of a finite index set I and contents $c_i \in \mathbb{H}$ with $c_i \geq 0$ and $Z = \sum_{i \in I} c_i > 0$. Define

$$\int_{\Omega} f := \frac{\sum_{i \in I} f(i) c_i}{Z}, \quad P_{\Omega}(E) := \int_{\Omega} \mathbf{1}_E \quad (E \subseteq I).$$

The ambient context Ω includes the partition and its contents.

Theorem 3.2 (Normalization and refinement). *The integral is linear, positive, and sends 1 to 1. The probability is finitely additive, lies between zero and one, and obeys $P_{\Omega}(E^c) = 1 - P_{\Omega}(E)$. Replacing each region by finitely many subregions whose contents sum to its content preserves integrals of functions constant on the original region.*

Proof. Expand the finite sums. Positivity uses $c_i \geq 0$ and $Z > 0$. Refinement follows by regrouping the subregion contents inside each original summand. No infinite interchange of sums is involved. $\square$

For $P_{\Omega}(E) > 0$, conditioning is explicitly a change of context:

$$\int_{\Omega|E} f = \frac{\int_{\Omega} \mathbf{1}_E f}{P_{\Omega}(E)}.$$

Ordinary restriction $\int_{\Omega} \mathbf{1}_E f$ does not change the normalizer. In a product of two contexts the contents multiply; finite algebra gives Fubini's identity and the usual factorization for independent coordinate events. Independence is a modeling assumption, not a consequence of rotation invariance.

4 Geometry, calibration, and endpoint corrections

Suppose the real coefficient $F(v)$ assigned to a segment displacement is rotation invariant, satisfies $F(tv) = tF(v)$ for $t \geq 0$, and $F(1, 0) = 1$. Then

$$F(x, y) = \sqrt{x^2 + y^2}.$$

Indeed, for positive length ℓ , the coefficients $x/\ell, y/\ell$ define an orthogonal rotation taking $(1, 0)$ to $(x/\ell, y/\ell)$. Scaling proves the claim; the zero case follows by scaling by zero. This characterizes one coefficient, not every possible probability rule.

Fix planar point content ε^2 . If closed segments have the graded form $\ell\varepsilon + k$, with the same endpoint coefficient k , joining two at one endpoint and applying inclusion–exclusion gives $k = 2k - \varepsilon^2$. Thus

$$\nu_{\varepsilon}([a, b]) = \ell\varepsilon + \varepsilon^2, \quad \nu_{\varepsilon}([a, b)) = \ell\varepsilon, \quad \nu_{\varepsilon}((a, b)) = \ell\varepsilon - \varepsilon^2. \quad (1)$$

The latter two expressions concern positive standard lengths. A degenerate closed segment is a point. These corrections are needed for exact finite additivity, even when omitted from a leading-order display.

For a closed convex polygon of area A and perimeter P , take

$$\nu_\varepsilon(K) = A + \frac{P}{2}\varepsilon + \varepsilon^2. \quad (2)$$

An internal cut of length L contributes twice to the sum of the two perimeters. Subtracting the shared closed segment cancels that excess and one point-level term. This checks the elementary dissection identity; it is not a formal construction of a valuation on all geometric sets.

4.1 The Wills polynomial connection

For a compact convex $K \subseteq \mathbb{R}^d$, let $V_j(K)$ denote its intrinsic volumes in the standard normalization. The Wills polynomial is $g_K(z) = \sum_j V_j(K)z^j$ [3]. Our proposed content is precisely

$$\nu_\varepsilon(K) = \sum_{j=0}^d V_j(K)\varepsilon^{d-j} = \varepsilon^d g_K(\varepsilon^{-1}). \quad (3)$$

This is formal polynomial evaluation, not a dilation of a standard set by a nonstandard scalar. Thus the coefficient polynomial is established geometry; the normalization and precision interface are our subject. For general event algebras, a compatible extension and positivity proof are still necessary. A valuation on convex sets alone is not yet a probability on all subsets of a board.

5 Square, cube, and coordinate codimension

For a closed rectangle, equation (2) factors as $\nu_\varepsilon([0, a] \times [0, b]) = (a + \varepsilon)(b + \varepsilon)$. The selected orthogonal-product rule therefore gives

$$\nu_\varepsilon([0, 1]^d) = (1 + \varepsilon)^d.$$

Set $\eta = \varepsilon/(1 + \varepsilon)$; this is a change of gauge, not an equality between η and ε .

Theorem 5.1 (Coordinate codimension). *If r coordinates in a closed unit d -cube are fixed and the other $d - r$ range over the entire unit interval, the product context gives*

$$P(E) = \frac{(1 + \varepsilon)^{d-r} \varepsilon^r}{(1 + \varepsilon)^d} = \eta^r.$$

Proof. A free coordinate contributes $1 + \varepsilon$; a fixed coordinate contributes ε . Multiply, then divide by the whole-cube content. $\square$

In particular, the cube, a full coordinate slice, a full axis crossing, and a point have probabilities $1, \eta, \eta^2, \eta^3$. A general closed segment of length L in the square has

$$P(L) = L\eta + (1 - L)\eta^2,$$

so its diagonal has $\sqrt{2}\eta + (1 - \sqrt{2})\eta^2$. The corresponding spatial segment has $L\eta^2 + (1 - L)\eta^3$. The length coefficient is geometric; the correction records its endpoints.

These statements differ from a coordinate-grid sampling convention, where a diagonal and an axis row can contain the same number of samples. That is a different context, not a contradiction with Euclidean length.

6 Disk and solid-ball darts

The standard intrinsic-volume normalization is

$$\text{vol}(K + tB_d) = \sum_{j=0}^d \kappa_{d-j} V_j(K) t^{d-j}, \quad (\kappa_0, \kappa_1, \kappa_2, \kappa_3) = (1, 2, \pi, 4\pi/3),$$

as in [4]. For round bodies, expanding the ordinary radius polynomials $\pi(r+t)^2$ and $(4\pi/3)(r+t)^3$ and matching coefficients gives the selected contents

$$\nu_\varepsilon(D_r) = \pi r^2 + \pi r\varepsilon + \varepsilon^2, \quad \nu_\varepsilon(B_r) = \frac{4\pi r^3}{3} + 2\pi r^2\varepsilon + 4r\varepsilon^2 + \varepsilon^3.$$

This step uses given real geometric calibrations and finite polynomial arithmetic. It does not derive ordinary area or π from field axioms. For radius one, write $Z_D = \nu_\varepsilon(D_1)$ and $Z_B = \nu_\varepsilon(B_1)$.

Board	Event	Exact probability
Disk	point	ε^2/Z_D
Disk	diameter	$(2\varepsilon + \varepsilon^2)/Z_D$
Ball	point	ε^3/Z_B
Ball	diameter	$(2\varepsilon^2 + \varepsilon^3)/Z_B$
Ball	central disk	$(\pi\varepsilon + \pi\varepsilon^2 + \varepsilon^3)/Z_B$

These formulas have concrete positive contexts. For the disk, the successive disjoint regions are a point, the rest of its diameter, and the rest of the board, with contents

$$\varepsilon^2, \quad 2\varepsilon, \quad \pi + (\pi - 2)\varepsilon.$$

For the ball, the nested point, diameter, and central disk yield the four disjoint-region contents

$$\varepsilon^3, \quad 2\varepsilon^2, \quad \pi\varepsilon + (\pi - 2)\varepsilon^2, \quad \frac{4\pi}{3} + \pi\varepsilon + (4 - \pi)\varepsilon^2.$$

They are positive and sum to Z_D and Z_B , respectively, using $2 < \pi < 4$. The normalized-context theorem therefore applies. In either board a point on the chosen diameter has the exact conditional probability

$$P(\text{point} \mid \text{diameter}) = \frac{\varepsilon}{2 + \varepsilon}. \quad (4)$$

The denominator of the whole board and its transverse powers of ε cancel. “Ball” here means the solid three-dimensional region; sampling only a spherical surface requires a different, two-dimensional context.

7 Algebraic precision and normalized integration

Definition 7.1. For a fixed positive infinitesimal gauge g and $n \in \mathbb{Z}$, write

$$r \in \mathcal{O}(g^n) \iff \exists M \in \mathbb{R}_{\geq 0} : |r| \leq Mg^n.$$

The statement $x = y + \mathcal{O}(g^n)$ means $x - y \in \mathcal{O}(g^n)$.

The bounding coefficient must be standard real. Equivalently, $\mathcal{O}(g^n) = g^n \mathcal{F}$, where $\mathcal{F}$ is the subring of elements bounded by standard reals. There is no limiting variable. Integer orders also accommodate infinite scales. For $0 < g \leq 1$, the elementary rules are

$$\mathcal{O}(g^m)\mathcal{O}(g^n) = \mathcal{O}(g^{m+n}), \quad \mathcal{O}(g^n)/g^m = \mathcal{O}(g^{n-m}), \quad \mathcal{O}(g^n) \subseteq \mathcal{O}(g^m) \ (m \leq n).$$

Addition uses the weaker of the two orders. Reciprocals preserve an absolute error order when both reciprocals are bounded; an infinitesimal denominator requires explicit scale tracking.

Proposition 7.2 (Uniform precision under normalization). *If $|f(i) - h(i)| \leq Mg^n$ for every region of a positive normalized context, using one standard real $M \geq 0$, then*

$$\int_{\Omega} f = \int_{\Omega} h + \mathcal{O}(g^n).$$

Proof. Integrate the inequalities $-Mg^n \leq f - h \leq Mg^n$. Positivity, linearity, and $\int_{\Omega} 1 = 1$ give the required two-sided bound. $\square$

For example, a spatial segment has $P(L) = L\eta^2 + \mathcal{O}(\eta^3)$, with exact expression $L\eta^2 + (1 - L)\eta^3$ retained. A closed planar patch of area A and perimeter P in the unit cube has

$$P(\text{patch}) = A\eta + (P/2 - 2A)\eta^2 + (A - P/2 + 1)\eta^3 = A\eta + \mathcal{O}(\eta^2).$$

An $\mathcal{O}$ statement by itself forgets the omitted coefficients. More precision must come from the retained exact expression or additional information. Cancellation cannot justify dropping a remainder: $(1 + g) - 1 = g$, not zero. The exact identity

$$(1 + g)^{-1} = 1 - g + \frac{g^2}{1 + g} = 1 - g + \mathcal{O}(g^2)$$

illustrates finite algebraic expansion without an infinite series.

8 Points, dots, halos, and the event domain

A singleton, a radius- ε dot, and a halo are distinct sets. For example, displacement $\varepsilon/2$ is in the dot but is not its center; displacement 2ε lies in the halo but not the dot. Here a halo contains points at every infinitesimal displacement, not merely standard real multiples of one chosen ε .

The normalized integral can give a singleton positive content directly. For any admissible support E with $P_{\Omega}(E) > 0$, the density $\delta_E = \mathbf{1}_E/P_{\Omega}(E)$ integrates to one. Its integral against f is the conditional average on E ; exact evaluation at a point requires singleton support (or an integrand constant on that support). This distinguishes point evaluation from dot averaging without silently assigning content to an entire halo.

The geometry is deliberately restricted to standard-coordinate data. Substituting an infinitesimal length into the open-segment formula would give, for example,

$$(\varepsilon/2)\varepsilon - \varepsilon^2 = -\varepsilon^2/2 < 0.$$

Consequently that formula cannot define a positive content on all infinitesimal-coordinate regions. A literal infinitesimal dot needs a separate compatible event construction. This obstruction is part of the theory's boundary, not a term to discard in an approximation.

9 Formalization and remaining mathematical work

The companion code separates exact rational-function arithmetic from the historical list backend. The current audited path uses the following modules:

- `Hyper/OrderedRational.lean`: field order and infinitesimal bounds.
- `Hyper/ContextIntegral.lean`: finite normalized integrals, conditioning, refinements, products, and support densities.
- `Hyper/GeometricContent.lean`: segment coefficient characterization, endpoint forcing, planar dissection algebra, and square formulas.
- `Hyper/CubicContent.lean`: box products, coordinate codimension, and positive cube contexts.
- `Hyper/RoundContent.lean`: selected positive disk/ball contexts and the exact ratios in equation (4).
- `Hyper/AlgebraicOrder.lean`: remainder arithmetic, uniform integral bounds, and selected geometric precision statements.
- `Hyper/AlgebraicSupport.lean`: point, dot, and halo distinctions.

Claim	Status in the companion
Exact field and finite normalized-context laws	Lean-checked
Square/cube/disk/ball selected event contexts	Lean-checked
Endpoint, product, and dissection identities	Lean-checked
Algebraic remainder and uniform integration rules	Lean-checked
Global geometric valuation independent of decomposition	Not implemented
All curved or infinitesimal-coordinate event sets	Not established
Full transfer, countable additivity, LLN or CLT	Not claimed
Novelty of the synthesis	Not established

The important next mathematical step is to specify one geometric event algebra and construct its compatible content, proving positivity and independence of decomposition. Local dissection identities and several finite contexts are evidence for such a development, but do not replace it. Uniqueness or minimal-axiom claims would require a separate theorem with all calibration and domain hypotheses stated.

Reproducibility and manuscript status

The implementation is available at <https://github.com/pannous/hyper-lean>; the code revision reviewed for this draft is 0cfcc3d. Run `bash test.sh` with the pinned Lean/mathlib setup. The script builds before testing and audits the named exact-path theorems. Their reported dependencies are `propext`, `Classical.choice`, and `Quot.sound`; no `sorryAx` or legacy raw-list equality axiom is accepted on that path. This is a theorem audit, not a claim that all historical modules are free of unfinished proofs.

AI tools assisted drafting and proof scripting. This draft requires human mathematical review and a fuller comparison with prior literature before a novelty claim or journal submission. It

supersedes the probability claims of the older introductory manuscript; it does not validate that manuscript's broader calculus and transfer claims.

References

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- [4] M. B. McCoy and J. A. Tropp, *From Steiner formulas for cones to concentration of intrinsic volumes*, equation (1.1), <https://arxiv.org/abs/1308.5265>.